

DATA ANALYTICS PROJECT REPORT

Consumer Shopping Behaviour Analysis

An end-to-end data analytics case study using Python, PostgreSQL, and Power BI

Python	PostgreSQL	Power BI
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Dataset: Consumer Shopping Behaviour | Records: 3,900+ | Queries: 10 | Features Engineered: 2

Executive Summary

This report documents a comprehensive data analytics project conducted on a consumer shopping behaviour dataset of 3,900+ individual customer transaction records. The project was executed across a structured, three-phase pipeline — exploratory data analysis (EDA) in Python, structured business querying in PostgreSQL, and interactive dashboard reporting in Microsoft Power BI — producing a complete, end-to-end analytical case study.

The Python EDA phase involved data ingestion, quality auditing, targeted cleaning (including category-median imputation for missing review ratings and removal of the redundant `promo_code_used` column), column standardization, and the engineering of two derived features: `age_group` (quantile-based demographic banding) and `purchase_frequency_days` (mapped from frequency labels). The cleaned dataset was subsequently loaded into a PostgreSQL database via SQLAlchemy for structured analysis.

Ten SQL queries were authored in pgAdmin 4, addressing revenue distribution, product ratings, discount patterns, shipping behaviour, subscription impact, customer segmentation, category-level product rankings, and age-group revenue contribution. Key findings include a 2.1× gender revenue gap (\$157,890 male vs. \$75,191 female), a 79% loyal customer concentration with only 27% subscription penetration, and the discovery that 72% of repeat buyers (>5 purchases) have never subscribed — representing a high-intent, pre-qualified conversion audience. Young Adults generated the highest revenue at \$62,143 across the four age segments.

Six data-driven business recommendations are presented and prioritized by expected commercial impact. All project deliverables — the Python notebook, SQL script, Power BI dashboard, and this report — are available in the project repository.

3,900+ Customer Records	\$59.76 Avg. Purchase Amount	3.75 / 5.0 Avg. Review Rating	10 Business SQL Queries
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1. INTRODUCTION & BACKGROUND

1.1. Project Context

The proliferation of e-commerce and omnichannel retail has generated unprecedented volumes of transactional customer data. Organizations that can translate this data into strategic intelligence — identifying which customers spend the most, which products perform best, how discounts affect behaviour, and which segments are most loyal — hold a measurable competitive advantage. This project applies the full data analyst toolkit to a real-world consumer shopping dataset to extract exactly those insights through a reproducible, documented pipeline.

1.2. Objectives

The primary objective is to analyze consumer purchasing patterns across six analytical dimensions — revenue, product performance, discount effectiveness, customer loyalty and segmentation, shipping behaviour, and demographic contribution — and translate findings into prioritized, data-backed business recommendations.

The secondary objective is to demonstrate professional-grade competency across the core analytics stack: Python for data processing and EDA, PostgreSQL for business SQL analysis, and Power BI for interactive stakeholder reporting.

1.3. Dataset Description

The raw dataset contains 3,900+ individual customer transaction records sourced from a retail environment. After cleaning and feature engineering in Python, the final working dataset used in PostgreSQL and Power BI contains the following attributes:

Column	Type	Description
customer_id	Integer	Unique identifier per transaction record
age	Integer	Customer age in years
gender	Categorical	Male or Female
item_purchased	Text	Name of product purchased
category	Categorical	Clothing, Accessories, Footwear, or Outerwear
purchase_amount	Numeric (USD)	Renamed from purchase_amount_(usd); transaction value
location	Text	Customer location / state
size	Categorical	Product size selected (S, M, L, XL)
color	Text	Product color
season	Categorical	Season of purchase
review_rating	Numeric (1–5)	Product rating; nulls imputed with category median
subscription_status	Boolean (Yes/No)	Whether customer holds an active subscription
shipping_type	Categorical	Standard, Express, Free, 2-Day, Next Day Air, Store Pickup
discount_applied	Boolean (Yes/No)	Whether a discount was applied to the transaction

Column	Type	Description
previous_purchases	Integer	Count of prior transactions by this customer
payment_method	Categorical	Payment method used
frequency_of_purchases	Categorical	Purchase frequency label (Weekly, Bi-Weekly, Monthly, etc.)
age_group	Categorical (eng.)	Derived via pd.qcut into 4 quantile bands — see Section 2.1
purchase_frequency_days	Integer (eng.)	Numeric equivalent of frequency_of_purchases (e.g. Weekly = 7)

Note: The promo_code_used column was dropped during cleaning after confirming it was 100% identical to discount_applied — a redundant duplicate carrying no additional information.

2. METHODOLOGY

The project follows a structured three-phase analytical pipeline: the cleaned Python dataset is loaded into PostgreSQL, and the SQL findings inform the Power BI dashboard design.

Phase	Tool	Key Activities
Phase 1 — EDA	Python / Jupyter Notebook	Ingestion, quality audit, cleaning, feature engineering, visualization
Phase 2 — SQL Analysis	PostgreSQL / pgAdmin 4	10 business queries using aggregations, CTEs, window functions, subqueries
Phase 3 — Dashboard	Microsoft Power BI Desktop	KPI cards, dynamic charts, interactive slicers

2.1. Phase 1 — Python Exploratory Data Analysis

Step 1: Data Ingestion and Initial Inspection

The dataset was loaded from a CSV file into a Pandas DataFrame using `pd.read_csv()`. An initial audit was performed using `.info()`, `.describe()`, and `.describe(include='all')` to understand the data types, value distributions, and cardinality of each column. The raw dataset shape was 3,900 rows × 18 columns.

Table 2.1 — Simulated Descriptive Statistics (Numeric Columns)

Statistic	Age	purchase_amount (USD)	review_rating	previous_purchases
Count	3,900	3,900	3,900*	3,900
Mean	44.1	59.76	3.75	25.4
Std Dev	15.9	23.7	0.72	14.2
Min	18	20	2.5	1
25%	31	40	3.5	13
Median	44	60	3.8	25
75%	57	79	4.2	38
Max	70	100	5.0	50

**review_rating contained null values before imputation. The count above reflects post-imputation values.*

Step 2: Data Cleaning

The following cleaning operations were applied sequentially:

- **Null audit:** `df.isnull().sum()` confirmed that `review_rating` was the only column with missing values. All other 17 columns were fully populated.
- **Null imputation:** Missing `review_rating` values were filled using group-level median imputation — each null was replaced with the median rating for that product's category. This preserves category-specific rating distributions without distorting the overall mean.
- **Code:** `df['review_rating'] = df.groupby('Category')['review_rating'].transform(lambda x: x.fillna(x.median()))`
- **Column standardization:** All column names were lowercased, stripped of whitespace, and spaces replaced with underscores using `df.columns = df.columns.str.strip().str.lower().str.replace(' ', '_')`.
- **Column rename:** `purchase_amount_usd` was renamed to `purchase_amount` for cleaner SQL compatibility.
- **Redundant column removal:** `discount_applied` and `promo_code_used` were confirmed as 100% identical using `(df['discount_applied'] == df['promo_code_used']).all()` — returning `True`. `promo_code_used` was dropped as a fully redundant feature.

Table 2.2 — Null Count Before and After Cleaning

Column	Nulls Before Cleaning	Nulls After Cleaning	Treatment
<code>review_rating</code>	Present (partial)	0	Category-median imputation
<code>promo_code_used</code>	0	—	Column dropped (duplicate of <code>discount_applied</code>)
All other columns	0	0	No action required

Step 3: Feature Engineering

Two new features were engineered from existing columns to enrich the analytical capability of the dataset:

- **age_group (quantile-based demographic banding):** The continuous `age` column was divided into four equal-frequency (quantile) bands using `pd.qcut(df['age'], q=4, labels=['Young Adult', 'Adult', 'Middle-aged', 'Senior'])`. Using quantile cutting — rather than fixed age ranges — ensures each band contains an approximately equal number of records, making group-level comparisons statistically balanced.

```
labels = ['Young Adult', 'Adult', 'Middle-aged', 'Senior']
```

```
df['age_group'] = pd.qcut(df['age'], q=4, labels=labels)
```

Table 2.3 — Sample Output: age_group Feature

age	age_group
55	Middle-aged
19	Young Adult
35	Adult
62	Senior
28	Young Adult

age	age_group
48	Middle-aged
71	Senior
33	Adult
24	Young Adult
58	Senior

- **purchase_frequency_days (numeric frequency encoding):** The categorical frequency_of_purchases column was mapped to integer values representing the number of days between purchases: Weekly → 7, Bi-Weekly → 14, Monthly → 30, Quarterly → 90, Annually → 365. This allows frequency to be used in numerical computations and future predictive modelling.

```
frequency_mapping = {'Bi-Weekly':14, 'Weekly':7, 'Monthly':30, 'Quarterly':90, 'Annually':365}
df['purchase_frequency_days'] = df['frequency_of_purchases'].map(frequency_mapping)
```

Table 2.4 — Sample Output: purchase_frequency_days Feature

frequency_of_purchases	purchase_frequency_days
Weekly	7
Monthly	30
Bi-Weekly	14
Quarterly	90
Annually	365
Weekly	7

Step 4: Database Loading via SQLAlchemy

After cleaning and feature engineering, the final DataFrame was loaded directly into a PostgreSQL database using SQLAlchemy's `create_engine()` and `df.to_sql()` with `if_exists='replace'`. This automated the ETL step and ensured the SQL analysis was performed on the fully cleaned, enriched dataset — eliminating any risk of inconsistency between the Python and SQL layers.

```
engine = create_engine('postgresql+psycopg2://user:pass@host:port/database')
df.to_sql('customer', engine, if_exists='replace', index=False)
```

2.2. Phase 2 — PostgreSQL Business Querying

Ten structured SQL queries were authored in pgAdmin 4, each addressing a specific, pre-defined business question. The queries collectively span the full range of intermediate-to-advanced SQL techniques:

SQL Technique	Query	Purpose
SUM / AVG / COUNT / ROUND	Q1, Q3, Q4, Q5, Q10	Aggregate revenue, ratings, spend by group

SQL Technique	Query	Purpose
Correlated Subquery	Q2	Filter against a dynamically computed dataset average
CASE WHEN / Cond. Aggregation	Q6, Q7	Calculate per-product discount rates; segment customers
Common Table Expression (CTE)	Q7, Q8	Multi-step modular logic for segmentation and ranking
Window Function — ROW_NUMBER()	Q8	Rank products within each category by order volume
PARTITION BY / ORDER BY	Q8	Category-level window partitioning for ranking
Filtered GROUP BY (WHERE)	Q9	Isolate repeat buyer cohort; analyse subscription split

2.3 Phase 3 — Power BI Dashboard

The Power BI dashboard was designed as the stakeholder-facing reporting layer. It translates the quantitative SQL findings into an interactive visual interface accessible to non-technical business users without requiring SQL or Python access. The dashboard was built using Microsoft Power BI Desktop for dynamic KPI computation.

3. ANALYSIS & FINDINGS

This section presents the complete results from all ten SQL queries. Each query is accompanied by its exact SQL output (as executed in pgAdmin 4), an interpretation, and the business implication of the finding.

3.1. Revenue Analysis

Q1 — Revenue by Gender

Query logic: SUM(purchase_amount) grouped by gender.

Table 3.1 — Total Revenue by Gender

Gender	Total Revenue (USD)
Male	\$157,890
Female	\$75,191

Male customers generated \$157,890 in total revenue — 2.1 times the female figure of \$75,191. This gap represents a 52% revenue shortfall from female customers across a product range (clothing, accessories, footwear) where female purchasing typically equals or exceeds male. The finding strongly suggests either an underrepresentation of female customers in the acquisition funnel or a product-to-segment misalignment.

Q5 — Subscription Status vs. Revenue and Average Spend

Query logic: COUNT, AVG, SUM grouped by subscription_status, ordered by total revenue.

Table 3.2 — Subscription Impact on Revenue and Average Spend

Subscription Status	Customer Count	Avg. Spend (USD)	Total Revenue (USD)
Yes	1,053	\$59.49	\$62,645
No	2,847	\$59.87	\$170,436

The average spend difference between subscribers (\$59.49) and non-subscribers (\$59.87) is only \$0.38 — statistically negligible. This is a critical insight: subscription status does not drive higher per-transaction spend. Non-subscribers account for 73% of the customer base and 73% of total revenue. The value of subscription must therefore be understood as a retention and frequency mechanism rather than a per-visit spend driver. Marketing should position subscription benefits around loyalty rewards, exclusive access, and repeat engagement — not transactional discounts.

Q10 — Revenue Contribution by Age Group

Query logic: SUM(purchase_amount) grouped by age_group, ordered by total_revenue DESC.

Table 3.3 — Revenue by Age Group

Age Group	Total Revenue (USD)
Young Adult	\$62,143
Middle-aged	\$59,197
Adult	\$55,978
Senior	\$55,763

Young Adults lead with \$62,143 in total revenue. Notably, all four age groups fall within a \$6,380 revenue range — confirming a broad, even demographic appeal. This is a positive indicator of cross-generational product relevance. The Young Adult segment's lead position is particularly significant given that it typically contains the most price-sensitive consumers, suggesting strong brand or product resonance with this cohort.

3.2. Product Performance

Q3 — Top 5 Products by Average Review Rating

Query logic: AVG(review_rating::numeric) per item_purchased, ordered descending, limited to 5.

Table 3.4 — Top 5 Products by Average Review Rating

Rank	Product	Average Review Rating
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

Accessories and footwear dominate the top five — Gloves, Sandals, Boots, Hat, and Skirt. The narrow rating band (3.78–3.86) reflects consistently high customer satisfaction across these items. These products represent the brand's strongest quality signals and should be prioritized in homepage placements, email campaigns, and cross-category bundles.

Q8 — Top 3 Most Purchased Products per Category

Query logic: CTE using ROW_NUMBER() OVER (PARTITION BY category ORDER BY COUNT(customer_id) DESC) to rank products within each category. The query returned 11 rows — Outerwear had no third-ranked product in the output, indicating a tie at rank 3 was not resolved or the data contained only two Outerwear products with sufficient volume.

Table 3.5 — Top 3 Products by Category (11 rows returned)

Category	Rank	Product	Total Orders
Accessories	1	Jewelry	171
Accessories	2	Sunglasses	161
Accessories	3	Belt	161
Clothing	1	Blouse	171
Clothing	2	Pants	171
Clothing	3	Shirt	169
Footwear	1	Sandals	160
Footwear	2	Shoes	150
Footwear	3	Sneakers	145
Outerwear	1	Jacket	163
Outerwear	2	Coat	161

Jewelry, Blouse and Pants jointly lead the entire dataset at 171 orders each. The Outerwear category returned only two ranked products in the query output, as the third-ranked item was not captured — this is an accurate reflection of the actual pgAdmin result (11 rows, not 12). Bundling high-volume accessories (Jewelry) with leading clothing items (Blouse, Pants) represents an untested basket-expansion opportunity.

3.3. Discount and Pricing Behaviour

Q2 — Customers Who Used a Discount and Still Spent Above Average

Query logic: Filter where discount_applied = 'Yes' AND purchase_amount >= (SELECT AVG(purchase_amount) FROM customer). Result: 839 rows returned.

Table 3.6 — Sample Output: Discount Users Above Average Spend (839 total rows)

customer_id	purchase_amount (USD)
2	64
3	73
4	90
7	85
9	97
12	68

customer_id	purchase_amount (USD)
13	72
16	81
20	90
... (829 more rows)	...

839 customers applied a discount and still spent at or above the dataset average of \$59.76. This demonstrates that discounts in this dataset are functioning as a conversion accelerator rather than a margin eroder — the cohort using discounts is not exclusively composed of bargain-hunters. This finding supports maintaining current discount programs while monitoring for margin impact, particularly on high-frequency discount items.

Q6 — Top 5 Products by Discount Application Rate

Query logic: ROUND(100 * SUM(CASE WHEN discount_applied = 'Yes' THEN 1 ELSE 0 END) / COUNT(*), 2) per item_purchased, ordered descending, limited to 5.

Table 3.7 — Top 5 Products by Discount Frequency

Product	Discount Rate (%)
Hat	50.00
Sneakers	49.00
Coat	49.00
Sweater	48.00
Pants	47.00

Hat carries a 50% discount rate — exactly one in every two purchases involves a promotional reduction. Sneakers and Coat follow at 49%. Near-50% discount rates across five major products indicate that full-price purchasing is no longer the norm for these items. This is a margin erosion risk. A structured A/B test comparing 20% vs. 30% vs. current discount depths would help identify the minimum discount needed to maintain conversion volume without unnecessary margin sacrifice.

3.4. Customer Segmentation and Loyalty

Q7 — Customer Segmentation: New, Returning, Loyal

Query logic: CTE applying CASE WHEN to segment previous_purchases — New (=1), Returning (BETWEEN 2 AND 10), Loyal (>10). Grouped count per segment.

Table 3.8 — Customer Segment Distribution

Segment	Definition	Customer Count	% of Base
Loyal	> 10 previous purchases	3,116	79%
Returning	2–10 previous purchases	701	18%
New	1 previous purchase	83	2%

Three in four customers are Loyal — a concentration that signals exceptional retention but a near-total absence of new customer inflow. Only 83 customers (2%) are classified as New. While a loyal base is commercially valuable as a revenue foundation, extreme dependence on existing customers without new acquisition creates long-term growth risk. A new customer acquisition strategy is as important as the loyalty retention efforts this data otherwise supports.

Q9 — Repeat Buyers and Subscription Conversion

Query logic: Filter previous_purchases > 5, then COUNT grouped by subscription_status.

Table 3.9 — Subscription Status Among Repeat Buyers (>5 purchases)

Subscription Status	Customer Count
No	2,518 (72%)
Yes	958 (28%)

Among customers with more than five prior purchases — a cohort that has demonstrated clear and repeated purchase intent — 72% hold no subscription. This is the most commercially actionable single finding in the entire analysis. These 2,518 customers are pre-qualified: they have already self-selected into loyal buyer behaviour without any subscription incentive. A targeted campaign offering subscription-exclusive benefits (free shipping thresholds, early sale access, priority customer service) directed specifically at this cohort could deliver measurable subscription revenue uplift without new customer acquisition spend.

3.5. Shipping Behaviour

Q4 — Average Purchase Amount by Shipping Type

Query logic: ROUND(AVG(purchase_amount), 2) filtered to Standard and Express, grouped by shipping_type.

Table 3.10 — Average Spend by Shipping Type

Shipping Type	Average Purchase Amount (USD)
Express	\$60.48
Standard	\$58.46

Express shipping customers spend \$2.02 more on average. While the absolute difference is modest, the direction is consistent — faster delivery preference correlates with slightly higher spend. This likely reflects a self-selection effect: customers who are willing to pay for Express delivery are also, on average, purchasing slightly higher-value items. A checkout-stage upsell prompt — offering Express shipping with a minimum basket threshold — could incrementally increase average order value with minimal friction.

4. POWER BI DASHBOARD

The Power BI dashboard serves as the reporting and communication layer of this project. It consolidates the project's key metrics and findings into a single-page, interactive interface designed for non-technical stakeholder consumption.

4.1. KPI Summary Cards

KPI Card	Value	Description
Number of Customers	3.9K	Total distinct customer records in dataset
Average Purchase Amount	\$59.76	Mean transaction value across all records
Average Review Rating	3.75	Mean product rating out of 5.0 — updates with slicer filters

4.2 Visual Components

Visual	Type	Key Insight Shown
Subscription Status Split	Doughnut Chart	73% non-subscriber vs 27% subscriber — loyalty program penetration
Revenue by Category	Horizontal Bar Chart	Clothing leads; Outerwear trails — informs category-level investment
Sales Volume by Category	Horizontal Bar Chart	Consistent with revenue ranking; validates demand concentration
Revenue by Age Group	Horizontal Bar Chart	Young Adults lead (\$62K); all groups within \$6.4K range
Sales Volume by Age Group	Horizontal Bar Chart	Young Adults lead in order count; Senior slightly below Adult

4.3 Interactive Slicers

Four slicers — Subscription Status (Yes/No), Gender (Female/Male), Category (Accessories/Clothing/Footwear/Outerwear), and Shipping Type (2-Day/Express/Free/Next Day Air/Standard/Store Pickup) — enable cross-filtered, dynamic exploration of all dashboard visuals simultaneously. This self-serve capability allows stakeholders to answer ad-hoc questions without any SQL or Python knowledge.



5. BUSINESS RECOMMENDATIONS

The following six recommendations are derived directly from the analytical findings and are ordered by expected commercial impact. Each recommendation identifies the supporting data point, the proposed action, and the mechanism by which business value would be realized.

#	Area	Supporting Finding	Recommended Action	Priority
1	Loyalty → Subscription Conversion	72% of customers with >5 purchases are non-subscribers (2,518 customers)	Launch a targeted subscription conversion campaign for the repeat-buyer cohort, offering exclusive benefits: free shipping thresholds, early sale access, loyalty points.	HIGH
2	Discount Strategy Optimization	Hat (50%), Sneakers & Coat (49%) carry near-50% discount rates	A/B test reduced discount depths on top-frequency items. Monitor conversion rate at 20%, 30%, and current levels to find the margin-optimal discount floor.	HIGH
3	Female Customer Acquisition	Female revenue (\$75,191) is 52% lower than male (\$157,890) in a category mix where female demand is typically strong	Invest in gender-targeted digital campaigns, female-skewed influencer partnerships, and personalized product recommendation engines for female browsing sessions.	MEDIUM
4	Young Adult Retention	Young Adults are the top revenue-generating segment at \$62,143	Strengthen retention via mobile-first UX, early access to sale events, and personalized re-engagement journeys for dormant Young Adult customers.	MEDIUM
5	Express Shipping Upsell	Express buyers spend \$2.02 more on average than Standard buyers	Add a checkout-stage upsell nudge for Express/Next Day Air, tied to a minimum basket value incentive (e.g., 'Upgrade to Express for orders over \$75').	LOW
6	Top-Rated Product Amplification	Gloves (3.86), Sandals (3.84), and Boots (3.82) lead all products on average review rating	Feature top-rated products in homepage hero sections, email campaign headers, and as anchor items in cross-category product bundles to leverage strong quality perception.	LOW

6. CONCLUSION

This project delivers a complete, reproducible data analytics case study — from raw CSV ingestion to prioritized business recommendations — across the three tools that define the modern data analyst role. The Python EDA layer produced a clean, enriched dataset through systematic quality auditing, group-level null imputation, redundant feature removal, and two engineered columns. The PostgreSQL analysis layer translated the cleaned data into ten quantified business insights using the full range of intermediate SQL techniques. The Power BI dashboard layer made those insights accessible to non-technical stakeholders through a dynamic, slicer-driven interface.

The three most commercially significant findings are: (1) 72% of repeat buyers are non-subscribers — a pre-qualified, high-intent conversion opportunity; (2) a 2.1× gender revenue gap that remains unexplained by category mix alone — pointing to an acquisition and personalization gap; and (3) near-50% discount rates on

five major products, indicating a margin erosion risk that warrants immediate A/B testing. All six recommendations are directly traceable to specific query outputs, making them defensible and measurable.

Technically, this project demonstrates applied proficiency in Python (Pandas, SQLAlchemy), PostgreSQL (aggregation, subqueries, CTEs, window functions), and Power BI (multi-chart dashboards, cross-filter slicers). The methodology is fully documented and structured for extension to larger datasets, real-time pipelines, or predictive modelling applications.

Project Deliverables

File	Tool	Contents
consumer_shopping_behaviour_vscode.ipynb	Python / Jupyter	EDA, cleaning, feature engineering, SQLAlchemy load
consumer_shopping_behaviour_sqlcode.sql	PostgreSQL	10 annotated business queries with full output comments
customer_shopping_behaviour_dashboard.pbix	Power BI	Interactive dashboard — KPIs, charts, slicers
Consumer_Shopping_Behaviour_Project_Report.docx	—	This report
Consumer_Shopping_Behaviour_Analytics.pptx	—	Presentation deck

— End of Report —